Denis Gruszkiewicz

Junior Software Engineer | Python, Rust, PostgreSQL, Linux, data pipelines

Wrocław | +48 511 592 887 | r3dflzp.ns@gmail.com

github.com/Flyingzaptop | antillia.tech

PROFILE

Junior engineer and student (Robotics and Process Automation at Politechnika Wrocławska; IT engineering at WSB Merito) with years of self-driven programming and data projects: streaming data collectors and Parquet/zstd storage with crash recovery in Python, an event-level order-book replay in Rust, PostgreSQL-backed workflow automation and a Django/React application. Over a year of work as a junior IT administrator in a hospital. Curious about databases, big data and how systems work.

RELEVANT SKILLS

- Python: asyncio/aiohttp collectors for eight exchange APIs, Polars/PyArrow/Parquet pipelines, append-only record log with zstd frames, checksums and tail recovery after failure (stargaze-l2-vwap).
- Rust: order-book replay and execution simulation, zstd-compressed DBN/Parquet handling with serde and clap (gc-mbo-research); Rust/PyO3 native extensions (stargazin). Learning level, project-based.
- PostgreSQL and SQL: SQL migrations, durable application state, basic queries (JobHunter, LabFlow).
- Linux and Git: Ubuntu deployment with systemd, Caddy and Gunicorn; Git/GitHub workflow; GitHub Actions CI with pytest, Node tests and CTest.
- Web: Django REST Framework, React/TypeScript, Node.js, SSE, REST/OpenAPI; Docker Compose configuration.
- Data and machine learning: PyTorch models, feature and label design, walk-forward validation, causality and leakage checks, scikit-learn; C++/CMake (graphics runtime project ARC).

EXPERIENCE

Junior IT Administrator | ZOZ w Bolesławcu | 06.2025 - present

- Day-to-day technical support for hospital staff; diagnosing hardware, operating system and application issues.
- Analysed user needs, tested scenarios, reproduced reported errors and fixed applications based on staff needs.
- Developed internal tools for repetitive tasks; worked on separate employee-management, scheduling and laboratory-process applications.

SELECTED PROJECTS

stargaze-l2-vwap - market data collectors | Python

Async connectors for eight exchanges, sequence handling and reconnects, append-only Parquet record log with recovery, desktop recorder and pytest CI.

gc-mbo-research - event-level data and Rust replay | Python, Rust

Order book reconstruction from CME MBO data, FIFO execution simulation, feature engineering and walk-forward evaluation; twenty experiment folders, current result NO_TRADE (negative results documented).

JobHunter - workflow automation | Node.js, n8n, PostgreSQL

Web data collection and analysis platform with Playwright, PostgreSQL migrations, local LLMs and CI regression tests.

EDUCATION

Politechnika Wrocławska | 10.2022 - present

Robotics and Process Automation; engineering studies in progress, five semesters completed.

WSB Merito we Wrocławiu | 03.2026 - present

Network, systems and IT security engineering; part-time engineering studies, second semester.

LANGUAGES AND AVAILABILITY

Based in Wrocław, Poland (CET). Flexible working hours; fully remote work preferred. From March 2027 onsite presence limited to 1–2 days per week if required. Open to part-time arrangements during studies; start date by agreement.